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随机过程期末复习

一. 相关理论与二阶矩过程

1. 自相关函数 $R_X(t,s) = E(X(t)X(s))$

① $R_X(t,t) \geq 0$ ② $R_X(t,s) = R_X(s,t)$

2. 宽平稳过程 W.S.S. ① $m_X(t) \equiv m$ ② $\forall T, R_X(t+T, s+T) = R_X(t,s) = R_X(t-s)$

令 $\tau = t-s$，考察 $R_X(\tau)$，有：① $R_X(0) \geq 0$ ② $R_X(\tau) = R_X(-\tau)$ ③ $|R_X(\tau)| \leq R_X(0)$

证③：LHS $= E(X(t)X(s))$，RHS $= \sqrt{R_X(0)} \cdot \sqrt{R_X(0)} = \sqrt{E(X^2(t))} \cdot \sqrt{E(X^2(s))}$，$\because E(XY)$ 是内积 $\therefore$ 证毕

(1) e.g. $X(t) = A\cos(2\pi f_0 t + \phi)$，$A, \phi$ r.v., independent，$\phi \sim U(0, 2\pi)$，证 $X(t)$ W.S.S.

证：(i) $E(X(t)) \overset{\text{i.d.}}{=\!=} E(A)E(\cos(2\pi f_0 t + \phi)) = E(A)\int_0^{2\pi} \cos(2\pi f_0 t + \phi)\frac{1}{2\pi} d\phi = 0$

(ii) $E(X(t)X(s)) = E(A^2)E\left(\frac{1}{2}[\underbrace{\cos(2\pi f_0(t+s)+2\phi)}_{=0} + \underbrace{\cos(2\pi f_0(t-s))}_{\text{非 r.v.}}]\right)$

$= \frac{1}{2}E(A^2)\cos(2\pi f_0(t-s))$

(2) e.g.2 $X(t) \in \{-1, 1\}$ 来回切换，切换次数 $\sim$ Poisson (λ)，求 $R_X(t,s)$

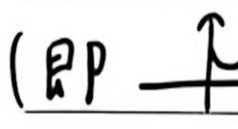
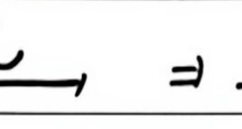
解：$R_X(t,s) = E(X(t)X(s)) \overset{\text{不独立}}{=\!=} 1 \cdot P(X(t)X(s)=1) + (-1)\cdot P(X(t)X(s)=-1)$

$P(X(t)X(s)=1) = P(N(t,s) \text{ is even}) = \sum_{k\in even} \frac{(\lambda(t-s))^k}{k!}\exp(-\lambda(t-s))$

$= \frac{1}{2}(e^{\lambda(t-s)} + e^{-\lambda(t-s)})\exp(-\lambda(t-s)) = \frac{1}{2}[1 + e^{-2\lambda(t-s)}]$

$\Rightarrow R_X(t,s) = \frac{1}{2}(1+e^{-2\lambda(t-s)}) - \frac{1}{2}(1-e^{-2\lambda(t-s)}) = \exp(-2\lambda(t-s))$，$\forall t<s$

(3) 命题：if $\exists T \neq 0$, $R_X(0) = R_X(T)$，则 $\forall \tau$ 有 $R_X(\tau) = R_X(\tau+T)$ (Periodic)

(即  $\Rightarrow$ ) 应用：  一定不是自相关函数

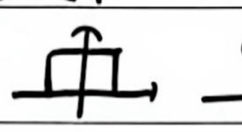
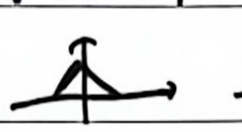
3. 函数正定：f is p.d. $\Leftrightarrow \forall n, \forall t_1, \cdots, t_n$，$(f(t_i - t_j))_{ij}$ is p.d.

(矩阵正定定义为 $A \in \mathbb{R}^{n\times n}$ is p.d. $\Leftrightarrow \forall x \in \mathbb{R}^n$, $x^TAx \geq 0$)

Thm：相关函数 $\Leftrightarrow$ 正定函数

相关的定义可扩充至 $\mathbb{C}$，$R_X(t,s) \triangleq E(X(t)X(s)^*)$，$R_X(t,s) = R_X(s,t)^*$

更透彻、常用的解读：f is p.d. $\Leftrightarrow \mathcal{F}\{f\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(t)\exp(-j\omega t)dt \geq 0$

应用：  $\xrightarrow{\mathcal{F}}$ Sa $\therefore$ ×；  $\xrightarrow{\mathcal{F}}$ Sa^2 $\therefore$ ✓；Gauss $\xrightarrow{\mathcal{F}}$ Gauss $\therefore$ ✓

4. $X(t)$ 一般不 s.t. 绝对可积，无法定义 FT $\hat{X}(\omega) = \int_{-\infty}^{\infty} x(t)\exp(-j\omega t)dt$

因此功率谱密度定义为 $S_X(\omega) = \lim_{T\to\infty} \frac{1}{T} E\left|\int_{-\frac{T}{2}}^{\frac{T}{2}} X(t)\exp(-j\omega t)dt\right|^2$

$\Rightarrow S_X(\omega) = \int_{-\infty}^{\infty} R_X(\tau)\exp(-j\omega\tau)d\tau$

$\updownarrow$

$R_X(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_X(\omega)\exp(j\omega\tau)d\omega$

KOKUYO

Thm: 若 $X(t) \in \mathbb{R}$，则 $S_X(\omega) = S_X(-\omega)$

证：$X(t) \in \mathbb{R} \Rightarrow R_X(\tau) = E(X(t)X(t+\tau)) \in \mathbb{R} \Rightarrow S_X(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} R_X(\tau)(\cos\omega\tau + j\sin\omega\tau)\,d\tau$

$\underline{\underline{R_X \text{ 偶}}}\ \int_{-\infty}^{\infty} R_X(\tau)\cos(\omega\tau)\,d\tau = S_X(-\omega)$ 证毕

5. Thm: $\xrightarrow[X(t)]{\text{w.s.s.}}$ $\boxed{\begin{matrix}\text{LTI}\\ h\end{matrix}}$ $\xrightarrow{Y(t)}$ 则 $Y(t)$ w.s.s. 且 $S_Y(\omega) = |H(\omega)|^2 S_X(\omega)$

e.g.1 $Y(t) = \frac{d}{dt}X(t)$ (注：$\hat{X}(\omega)$ 未 ∴ ~~$Y(\omega) = j\omega X(\omega)$~~)

解：$R_Y(t,s) = E(Y(t)Y(s)) = E\left(\frac{d}{dt}X(t)\frac{d}{ds}X(s)\right) = \frac{\partial^2}{\partial t\partial s}E(X(t)X(s))$

$= \frac{\partial^2}{\partial t\partial s}R_X(t-s) = -\frac{d^2}{d\tau^2}R_X(\tau) = R_Y(\tau) \Rightarrow S_Y(\omega) = \omega^2 S_X(\omega)$

$\Rightarrow |H(\omega)|^2 = |j\omega|^2$

(∴微分器：高通)

e.g.2 $Y(t) = \frac{1}{T}\int_{t-\frac{T}{2}}^{t+\frac{T}{2}} X(s)\,ds$ (∴积分器：低通)

解：$Y(t) = X(t) * h(t)$，$h(t) = \begin{cases}\frac{1}{T}, & -\frac{T}{2} < t < \frac{T}{2}\\ 0, & \text{else}\end{cases} \Rightarrow H(\omega) = Sa(\omega\frac{T}{2}) \Rightarrow S_Y(\omega) = S_X(\omega)Sa^2(\frac{\omega T}{2})$

e.g.3 (Brown Motion) $B(t)$ 不 w.s.s.，$E(B(t)) = 0$，$R_B(t,s) = \min(t,s)$，$Y(t) = \frac{d}{dt}B(t)$，求 $S_Y(\omega)$

(∵ $B(t)$ 不 w.s.s. ∴ 上述 Thm 不能用) $R_Y(t,s) = \frac{\partial^2}{\partial t\partial s}E(B(t)B(s)) = \frac{\partial^2}{\partial t\partial s}R_B(t,s)$

$= \frac{\partial^2}{\partial t\partial s}\min(t,s) = \frac{\partial^2}{\partial t\partial s}\frac{t+s-|t-s|}{2} = \delta(t-s)$，$S_Y(\omega) = \mathcal{F}\{R_Y(\tau)\} = 1$

(PS: $\frac{d}{dx}|x| = \text{sgn}(x)$，$\frac{d}{dx}\text{sgn}(x) = \frac{d}{dx}[U(x) - U(-x)] = \delta(x) + \delta(-x) = 2\delta(x)$)

6. 谱表示：$X(t) = \int_{-\infty}^{\infty} \exp(j\omega t)\,dF_X(\omega)$

$E(dF_X(\omega)\overline{dF_X(\omega')}) = \begin{cases}\frac{1}{2\pi}S_X(\omega)\,d\omega, & \omega' = \omega\\ 0, & \omega' \neq \omega\end{cases}$ (与 $R_X(\tau) \xleftrightarrow{\text{Fourier}} S_X(\omega)$ 互洽)

$dF_Y(\omega) = H(\omega)dF_X(\omega)$

7. 补充：互相关，互谱 (设 w.s.s.)

$R_{XY}(\tau) = E(X(t+\tau)\overline{Y(t)})$，$R_{YX}(\tau) = E(Y(t+\tau)\overline{X(t)})$

$R_{XY}(\tau) \longleftrightarrow S_{XY}(\omega)$

$R_{YX}(\tau) \longleftrightarrow S_{YX}(\omega)$

Thm: $\xrightarrow[X(t)]{\text{WSS}}$ $\boxed{\begin{matrix}\text{LTI}\\ h\end{matrix}}$ $\xrightarrow{Y(t)}$ 则 $S_{XY}(\omega) = \overline{H(\omega)}S_X(\omega)$，$S_{YX} = H(\omega)S_X(\omega)$

二. Gaussian Process

$X(t)$ Continous Time, Continous State

1. 定义: $f_X(x) = \frac{1}{(2\pi)^{\frac{n}{2}}(\det\Sigma)^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-u)^T\Sigma^{-1}(x-u)\right)$, $x \in \mathbb{R}^n$. $u = E(x) \in \mathbb{R}^n$

协方差阵 $\Sigma = E\left((x-u)(x-u)^T\right) \in \mathbb{R}^{n\times n}$

二维: $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$, $f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\left(\frac{x_1-u_1}{\sigma_1}\right)^2 + \left(\frac{x_2-u_2}{\sigma_2}\right)^2 - 2\rho\frac{x_1-u_1}{\sigma_1}\frac{x_2-u_2}{\sigma_2}\right)\right)$

注: 协方差 $Cov(X,Y) = E[(X-EX)(Y-EY)]$

相关系数 $Corr(X,Y)$ 或记作 $\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}$

2. 特征函数

$\phi_X(w) = E\left(\exp(jw^TX)\right) = \exp\left(jw^Tu - \frac{1}{2}w^T\Sigma w\right)$

3. Thm (Linear Invariance): $X \sim N(u,\Sigma)$, $Y = AX \Rightarrow Y \sim N(Au, A\Sigma A^T)$

推论 (Joint Gauss ⇒ Boundary Gauss): $X = (X_1,\cdots,X_n)^T \sim N \Rightarrow \widetilde{X} = (X_{n_1},\cdots,X_{n_k})^T \sim N$, $\forall\{n_1,\cdots,n_k\} \subseteq \{1,\cdots,n\}$

注意: Boundary Gauss $\not\Rightarrow$ Joint Gauss

e.g. $f(x_1,x_2) = \frac{1}{2\pi}\exp\left(\frac{x_1^2+x_2^2}{2}\right)(1+\sin x_1\sin x_2)$ 非 Gauss, 但 $\int f dx_{1,2} = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x_{1,2}^2}{2}\right)$ Gauss

Thm: 设 $X \in \mathbb{R}^n$, 则 $X \sim N \Leftrightarrow \forall \alpha \in \mathbb{R}^n$, $\alpha^TX \sim N$

补充: Gauss 过程 独立 $\Leftrightarrow$ 不相关

证 $\Leftarrow$: uncorrelated $\Leftrightarrow Cov(X,Y) = 0 \Rightarrow \rho = 0 \Rightarrow f_{X_1,X_2}(x_1,x_2) = \left(\frac{1}{\sqrt{2\pi}\sigma_1}e^{\frac{(x_1-u_1)^2}{2\sigma_1^2}}\right)\left(\frac{1}{\sqrt{2\pi}\sigma_2}e^{\frac{(x_2-u_2)^2}{2\sigma_2^2}}\right)$ $f_{X_1}(x_1)$ $f_{X_2}(x_2)$

4. 条件 Gauss

Thm: 设 $\begin{pmatrix}X_1\\X_2\end{pmatrix} \sim N\left(\begin{pmatrix}u_1\\u_2\end{pmatrix}, \begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix}\right)$, 则 $X_2|X_1 \sim N(u_{2|1}, \Sigma_{2|1})$

$u_{2|1} = u_2 + \Sigma_{21}\Sigma_{11}^{-1}(X_1-u_1)$, $\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_1^{-1}\Sigma_{12}$

5. Nonlinear Gauss

Price Theorem: 设 $(X_1,X_2) \sim N(0,0,\sigma_1,\sigma_2,\rho)$, $g(x_1,x_2)$ Nonlinear, 则

$$\frac{\partial}{\partial\rho}E(g(X_1,X_2)) = \sigma_1\sigma_2 E\left(\frac{\partial^2 g}{\partial X_1\partial X_2}\right)$$

e.g. $X(t)$ Gauss. W.S.S., $E(X(t)) = 0$, $Y(t) = sgn(X(t))$, 求 $R_Y(t,s)$

使用 Price Thm: $g(x_1,x_2) = sgn(x_1)sgn(x_2)$, $\frac{\partial^2 g}{\partial x_1\partial x_2} = 4\delta(x_1)\delta(x_2)$

$$\sigma_1\sigma_2 E\left(\frac{\partial^2 g}{\partial x_1\partial x_2}\right) = \sigma_1\sigma_2\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\iint_{\mathbb{R}^2}\exp(\cdots)\,4\delta(x_1)\delta(x_2)\,dx_1dx_2 = \frac{2}{\pi\sqrt{1-\rho^2}}$$

$\Rightarrow R_Y(t,s) = E(Y(t)Y(s)) = E(g(X(t),X(s))) = \frac{2}{\pi}\sin^{-1}(\rho) + C$ $\xrightarrow{\because R_Y(t,s)|_{\rho=0} = 0}$ $\therefore C = 0$ $\frac{2}{\pi}\sin^{-1}(\rho)$

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三. Poisson Process

$X(t)$ Continous time, Discrete State

1. 

点过程，计数过程

① Independent Increment $\forall t_1 < t_2 \le t_3 < t_4$, $X(t_4) - X(t_3)$ 与 $X(t_2) - X(t_1)$ 独立

② Stationary Increment $X(t_4) - X(t_1) \sim P(t_2 - t_1)$ 仅依赖时间长度

③ $P(X(\Delta t) \ge 2) / P(X(\Delta t) = 1) = o(\Delta t)$

工具：矩母函数 $G_X(z) = \underline{E(z^X)} = \underline{\sum_k p(X(t)=k) \cdot z^k}$

Characteristic Function ↔ z变换，处理 Discrete

相当于取 $z = e^{jw}$, ↔ FT，处理 Continous

推导得 $G_{X(t)}(z) = e^{(-\alpha + \alpha z)t} = e^{-\alpha t} \sum_{k=0}^{\infty} \frac{(\alpha t)^k}{k!} z^k = \sum_{k=0}^{\infty} P(X(t)=k) \cdot z^k$

$\Rightarrow P(X(t)=k) = \frac{(\alpha t)^k}{k!} \exp(-\alpha t)$, $P(X(t) - X(s) = k) = \frac{(\alpha(t-s))^k}{k!} \exp(-\alpha(t-s))$

$P(X(t_1)=k_1, X(t_2)=k_2, \cdots, X(t_n)=k_n) = P(X(t_1)=k_1) P(X(t_2-t_1)=k_2-k_1) \cdots P(X(t_n) - X(t_{n-1}) = k_n - k_{n-1})$

以下用 $N(t)$ 表示 Poisson Process，λ 表示其 Intensity

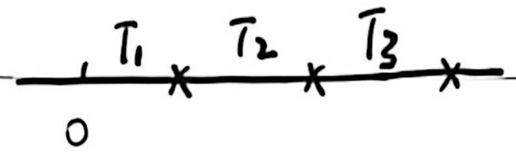
$E(N(t)) = \lambda t$, $Var(N(t)) = \lambda t$

设 $t > s$, $E(N(t)N(s)) = E((N(s) + N(t-s))N(s)) = \lambda s + (\lambda s)^2 + \lambda(t-s)\lambda s = \lambda s + \lambda^2 ts$ 不 W.S.S.

2. 设 $t > s$, 问 (1) $E(N(t) | N(s)) = E(N(t-s) + N(s) | N(s)) = \lambda(t-s) + N(s)$

(2) $E(N(s) | N(t))$: $P(N(s)=k | N(t)=n) = P(N(s)=k, N(t)=n) / P(N(t)=n) = P(N(s)=k, N(t-s)=n-k) / \cdots$

$= \frac{n!}{k!(n-k)!} (\frac{s}{t})^k (\frac{t-s}{t})^{n-k} = \binom{n}{k} (\frac{s}{t})^k (1 - \frac{s}{t})^{n-k} \Rightarrow N(s)=k | N(t)=n \sim B(n, \frac{s}{t})$ $(t > s)$

3. 

(1) $F_{T_1}(x) = P(T_1 \le x) = P(N(x) \ge 1) = 1 - P(N(x)=0) = 1 - \exp(-\lambda x)$

$f_{T_1}(x) = \lambda \exp(-\lambda x)$, $x \ge 0$ 指数分布！

$\Rightarrow$ 事件间隔 $\{T_k\}$ i.i.d $Exp(\lambda)$

(2) 第 n 次事件发生的时刻 $S_n = T_1 + \cdots + T_n$, $F_{S_n}(x) = P(S_n \le x) = P(N(x) \ge n) = \sum_{k=n}^{\infty} \frac{(\lambda x)^k}{k!} e^{-\lambda x}$

$\Rightarrow f_{S_n}(x) = \frac{d}{dx} F_{S_n}(x) = \frac{\lambda(\lambda x)^{n-1}}{(n-1)!} \exp(-\lambda x)$ Gamma 分布

4. 非齐次 Poisson

将 S.I. 改成 $\lim_{\Delta t \to 0} (1 - P(N(t+\Delta t) - N(t) = 0)) = N(t)$, 见 $\Rightarrow \begin{cases} \frac{d}{dt} G_{N(t)}(z) = (-\lambda(t) + \lambda(t) z) G_{N(t)}(z) \\ G_{N(0)}(z) = 1 \end{cases}$

$\Rightarrow G_{N(t)}(z) = \exp(\int_0^t \lambda(s) ds \, (z-1))$

$P(N(t)=k) = \frac{(\int_0^t \lambda(s) ds)^k}{k!} \exp(-\int_0^t \lambda(s) ds)$ Time Varied Intensity

$E(N(t)) = Var(N(t)) = \int_0^t \lambda(s) ds$

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5. Poisson 与二项的关系

(1) Poisson → Binomial : $P(N(s)=k \mid N(t)=n) \sim B(n, \frac{s}{t})$ $(s<t)$

(2) Binomial → Poisson: $\binom{n}{k} p^k (1-p)^{n-k} \xrightarrow{p\to 0,\ np\to\lambda,\ n\to\infty} \frac{\lambda^k}{k!}\exp(-\lambda)$

6. 复合 Poisson

设 $N(t)$ Poisson, $S(t)=\sum_{k=1}^{N(t)} X_k$ ("效果") (X_k r.v., i.i.d, independent of $N(t)$)

$\Rightarrow G_{S(t)}(z)=G_{N(t)}(G_{X_1}(z))=\exp(-\lambda t\,(G_{X_1}(z)-1))$, $E(S(t))=E(N(t))E(X_1)$

另一种视角：将假设 ① 改成 $P(N(\Delta t)=k \mid N(\Delta t)\ge 1) \xrightarrow{\Delta t\to 0} p_k$，即高阶项不忽略，亦导出上式

e.g.1 进门人数 ~ Poisson, 男女 $\begin{pmatrix} M & F \\ p & 1-p \end{pmatrix}$, 问进门的男同学人数 ~ ?

解: $S(t)=\sum_{k=1}^{N(t)} X_k$, $X_k=\begin{cases}1, & \text{Male } p\\ 0, & \text{Female } 1-p\end{cases}$ $\Rightarrow G_{S(t)}(z)=\exp(\lambda t(G_{X_1}(z)-1))=\exp(\lambda p t(z-1))$

e.g.2 $N_1(t)$, $N_2(t)$, λ_1, λ_2, independent, $N(t)=N_1(t)+N_2(t)$, $S(t)=N_1(t)-N_2(t)$

(1) $G_{N(t)}(z)=E(z^{N_1(t)+N_2(t)})=E(z^{N_1(t)})E(z^{N_2(t)})=e^{\lambda_1 t(z-1)}e^{\lambda_2 t(z-1)}=\exp((\lambda_1+\lambda_2)t(z-1))$

(2) $G_{S(t)}(z)=E(z^{N_1(t)})E(z^{-N_2(t)})=G_{N_1(t)}(z)\,G_{N_2(t)}(\frac{1}{z})=e^{\lambda_1 t(z-1)}e^{\lambda_2 t(\frac{1}{z}-1)}$

$=\exp\left((\lambda_1+\lambda_2)t\left(\frac{\lambda_1}{\lambda_1+\lambda_2}z+\frac{\lambda_2}{\lambda_1+\lambda_2}z^{-1}-1\right)\right)$

新强度 $G_{X_1}(z)$ 复合 Poisson $\quad X_1 \sim \begin{pmatrix} 1 & -1 \\ \frac{\lambda_1}{\lambda_1+\lambda_2} & \frac{\lambda_2}{\lambda_1+\lambda_2} \end{pmatrix}$

7. 顺序统计量

$(X_1,\cdots,X_n) \xrightarrow{i.i.d} f(x) \xrightarrow{\text{排序}} (X_{(1)}\le\cdots\le X_{(n)})$

① $F_{X_{(n)}}(x)=P(X_{(n)}\le x)=P(X_1\le x,\cdots,X_n\le x)=P(X_1\le x)\cdots P(X_n\le x)=F^n(x)$

② $F_{X_{(1)}}(x)=1-P(X_{(1)}>x)=1-P(X_1>x,\cdots,X_n>x)=1-P(X_1>x)\cdots P(X_n>x)=1-(1-F(x))^n$

③ $P(x\le X_{(k)}\le x+\Delta x)=\binom{n}{k-1}\binom{n-k+1}{n-k}F^{k-1}(x)(F(x+\Delta x)-F(x))(1-F(x+\Delta x))^{n-k}$

$\Rightarrow f_{X_{(k)}}(x)=\lim_{\Delta x\to 0}\frac{1}{\Delta x}P(x\le X_{(k)}\le x+\Delta x)=\binom{n}{k-1}\binom{n-k+1}{n-k}F^{k-1}(x)(1-F(x))^{n-k}f(x)$

④ $f_{X_{(k)},X_{(m)}}(x_k,x_m)=\binom{n}{k-1}\binom{n-k+1}{m-k+1}\binom{n-m+2}{n-m}\binom{2}{1}F^{k-1}(x_k)(F(x_m)-F(x_k))^{m-k-1}(1-F(x))^{n-m}f(x_k)f(x_m)$ $(k<m)$

$\Rightarrow f_{X_{(1)},\cdots,X_{(n)}}(x_1,\cdots,x_n)=\begin{cases} n!\prod_{k=1}^{n} f(x_k), & x_1\le x_2\le\cdots\le x_n \\ 0 & \text{else}\end{cases}$

8. 结论：$f_{S_1,\cdots,S_n\mid N(t)=n}(x_1,\cdots,x_n)$ 是 n 个独立的均匀分布的顺序统计量 $=\begin{cases}\frac{n!}{t^n}, & 0\le x_1\le\cdots\le x_n\le t\\ 0, & \text{else}\end{cases}$

其余内容见笔记

Thm: 若 $\forall\sigma$ Permutation, $g(x_1,\cdots,x_n)=g(x_{\sigma(1)},\cdots,x_{\sigma(n)})$, 则

$$\int_0^t\int_0^{x_n}\cdots\int_0^{x_2} g(x_1,\cdots,x_n)\,dx_1\cdots dx_n=\frac{1}{n!}\int_0^t\cdots\int_0^t g(x_1,\cdots,x_n)\,dx_1\cdots dx_n$$

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9. Filtered Poisson

$Y(t) = \sum_{k=1}^{N(t)} X_k(t, \tau_k)$ X_k i.i.d, X_k independent of $N(t)$. $(\tau_k \le t)$

则 $\phi_{Y(t)}(w) = \exp\left(\lambda \int_0^t [E(e^{jw X_1(t,\tau)}) - 1] d\tau\right)$

$E(Y(t)) = \frac{1}{j} \frac{\partial}{\partial w} \phi_{Y(t)}(w)\Big|_{w=0} = \lambda \int_0^t E(X_1(t,\tau)) d\tau$

四. Markov Chain

$X(t)$ Discrete time, Discrete states

1. 定义：$P(X_n = x_n | X_{n-1} = x_{n-1}, \cdots, X_1 = x_1) = P(X_n = x_n | X_{n-1} = x_{n-1})$，即 $P(C|BA) = P(C|B)$（未来 现在 过去）

Thm: $P(C|BA) = P(C|B) \Leftrightarrow P(CA|B) = P(C|B)P(A|B)$ 过去和未来地位等价

注：唯独当下一定不能模糊（$X_{n-1} = x_{n-1}$ 不能改成 $X_{n-1} \in$ 集合）

2. 转移概率 $P_{ij}(m,n) = P(X_n = x_j | X_m = x_i)$

平稳性：$\forall l$, $P_{ij}(m,n) = P_{ij}(m+l, n+l) = P_{ij}(n-m)$

3. Reachable 可达：$i \to j \Leftrightarrow \exists n, P_{ij}(n) > 0$

Communicate 相通：$i \leftrightarrow j \Leftrightarrow i \to j$ 且 $j \to i$

Closed set 闭集：$C \subseteq S \Leftrightarrow \forall i \in C, j \notin C, i \nrightarrow j$

Irreducible 不可约：S is irreducible $\Leftrightarrow$ S 无闭的真子集 $\Leftrightarrow$ S 所有状态都相通

4. 定义首达概率 $f_{ij}(n) = P(X_n = j, X_{n-1} \neq j, \cdots, X_1 \neq j | X_0 = i)$

性质：$\sum_{n=1}^{\infty} f_{ij}(n) \le 1$ （但是 $\sum_{n=0}^{\infty} P_{ij}(n)$ 可能 > 1）

$\exists$ 定义常返，i is Recurrent $\Leftrightarrow \sum_{n=0}^{\infty} f_{ii}(n) = 1$ $\Leftrightarrow \sum_{n=0}^{\infty} P_{ii}(n)$ 发散

i is Non-Recurrent $\Leftrightarrow \sum_{n=0}^{\infty} f_{ii}(n) < 1$ $\Leftrightarrow \sum_{n=0}^{\infty} P_{ii}(n)$ 收敛

5. 两大基本方程

$$\begin{cases} P_{ij}(n) = \sum_{k} P_{ik}(m) P_{kj}(n-m), \ \forall m & \text{空间维度 C-K 方程} \\ P_{ij}(n) = \sum_{k=1}^{n} f_{ij}(k) P_{jj}(n-k) & \text{时间维度 无名方程} \end{cases}$$

6. 命题 1：若 $i \leftrightarrow j$，则有：i Recurrent $\Leftrightarrow$ j Recurrent

命题 2：有限个状态的 Markov Chain 一定习 常返态.

$\xRightarrow{1+2}$ Thm：有限状态 + 不可约 $\Rightarrow$ 所有状态都常返 （正常返）

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7. 定义 $g_{ij}(n) = P(\#\{k: X_k = j \mid X_0 = i\} \geq n)$ 从 i 出发到 j 至少 n 次的概率

$\Rightarrow$ $g_{ij}(n+1) = \left(\sum_{m=1}^{\infty} f_{ij}(m)\right) g_{jj}(n)$ 第三方程

令 $j=i$ $\Rightarrow$ $g_{ii}(n+1) = f_{ii}\, g_{ii}(n)$ $\Rightarrow$ $g_{ii}(n) = f_{ii}^{\,n}$, $f_{ii} = \sum_{m=1}^{\infty} f_{ii}(m) \leq 1$

$\Rightarrow$ $g_{ii}(n) \xrightarrow{n\to\infty}$ { 1. $f_{ii}=1$ $\Leftrightarrow$ Recurrent
0. $f_{ii}<1$, $\Leftrightarrow$ Non-Recurrent

常返 $\Leftrightarrow$ 无穷多次返回的概率为1

8. 对常返的终极认识：若 i Recurrent，则 $i \to j \Rightarrow j \to i$

又由 $\{i \leftrightarrow j\} \Rightarrow \{i$ Recurrent $\Leftrightarrow j$ Recurrent$\}$，从而 i 常返 + $i \to j \Rightarrow j$ 常返

Recurrent 到哪哪就 Recurrent

9. 定义 Period 周期：$d_i = \text{GCD}\{k: P_{ii}(k) > 0\}$

$d_i = 1$ $\Leftrightarrow$ 非周期 Aperiodic

10. Ergodic Thm：不可约 + 非周期 $\Rightarrow$ $P_{ij}(n) \xrightarrow{n\to\infty} \pi_j$

Weak Ergodic Thm：j Recurrent + (包括) 不可约 $\Rightarrow$ $\frac{1}{n}\sum_{k=1}^{n} P_{ij}(k) \xrightarrow{n\to\infty} \frac{1}{a_j}$, $\forall i$

其中 $a_j = \sum_{k=1}^{\infty} f_{jj}(k)\cdot k = \begin{cases} <\infty & j \text{ 正常返 } (\frac{1}{a_j} > 0) \\ =\infty & j \text{ 零常返 } (\frac{1}{a_j} = 0) \end{cases}$ $\Rightarrow \frac{1}{a_j} = \pi_j$

11. $\pi = \pi P$ (一定有非零解，解为平稳分布，但只有极限分布存在时极限分布才 $= \pi$)

① $\pi = \pi P \Leftrightarrow \pi_j = \sum_i \pi_i P_{ij}$, $\forall j \Leftrightarrow \pi_i = \sum_j \pi_j P_{ji}$, $\forall i$ (只是把矩阵乘法算出来，目的是有时候不需要矩阵，这样更简单)

推论：若 $P(X_0 = i) = \pi_i$，则 $P(X_n = i) = P(X_0 = i) = \pi_i$, $\forall n \geq 1$

"平稳概率"

② $\pi = \pi P \Leftarrow \pi_i P_{ij} = \pi_j P_{ji}$ "细致平衡"

($i \to j$ 的流量 $= j \to i$ 的流量)

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过滤 Poisson

设 $N(t) \sim \text{Poisson}(\lambda t)$，第 k 次事件发生时刻为 S_k，令 $Y_k(t, S_k)$ 表示 S_k 发生下的第 k 次冲击在 t 时刻的贡献大。

Thm: 设 $Y(t) = \sum_{k=1}^{N(t)} Y_k(t, S_k)$，则 $E(Y(t)) = \lambda \int_0^t E(Y_1(t, s))\, ds$

公式成立条件说明：设 $Y_k(t, S_k) = D_k \cdot f(t, S_k)$，$D_k$ r.v. 为随机幅度，f 为确定函数

则要求 D_k i.i.d，且 D_k 与 $N(t)$ 独立。$E(Y_1(t, S))$ 是对 D 求期望，$= E(D) \cdot f(t, s)$。

e.g.1 (作业T14) 求 $E\left(\sum_{k=1}^{N(t)} e^{-(t-S_k)^2}\right)$ (相当于 $D_k \equiv 1$)

解：$Y(t) = \sum_{k=1}^{N(t)} Y_k(t, S_k)$，$Y_k(t, S_k) = e^{-(t-S_k)^2}$，$E(Y_1(t, S)) = e^{-(t-S)^2}$

$$\Rightarrow E(Y(t)) = \lambda \int_0^t e^{-(t-s)^2} ds = \lambda\sqrt{\pi} \int_0^{\sqrt{2}t} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = \lambda\sqrt{\pi}\left(\Phi(\sqrt{2}t) - 0.5\right)$$

e.g.2 (作业T16) $Y(t) = \sum_{k=1}^{N(t)} Y_k(t, S_k)$，$Y_k(t, S_k) = D_k \cdot \left(1 - \frac{t-S_k}{T_1}\right)_+$，$D_k \overset{\text{i.i.d}}{\sim} U(M_a, M_b)$

$$E(Y_1(t, S)) = E(D_k)\left(1 - \frac{t-S}{T_1}\right)_+ = \frac{M_a + M_b}{2}\left(1 - \frac{t-S}{T}\right)_+$$

$$E(Y(t)) = \lambda \frac{M_a + M_b}{2} \int_0^t \left(1 - \frac{t-s}{T}\right)_+ ds = \frac{\lambda(M_a + M_b)}{2} \int_{t-T_1}^t \left(1 - \frac{t-s}{T_1}\right) ds = \frac{\lambda(M_a + M_b)T}{4}$$

e.g.3 (23T3) 服务时间 $T \sim \text{Exp}(u)$，开门前 N 人等候，开门后 $N(t)$

解：$Y(t) = Y_1(t) + Y_2(t)$，$Y_2(t) = \sum_{k=1}^{N(t)} \delta_k(t, S_k)$，$\delta_k(t, S_k) = \begin{cases} 1 & t - S_k < T \\ 0 & t - S_k \geq T \end{cases}$

$$P(T > t) = \int_t^\infty u e^{-ux} dx = e^{-ut}$$

① $N = \text{const}$：$Y_1(t) \sim B(N, e^{-ut})$，$E(Y_1(t)) = Ne^{-ut}$

$$\phi_{Y_1(t)}(w) = \exp\left(\lambda \int_0^t \left[E\left(e^{jw\delta_1(t,s)} - 1\right)\right] ds\right)$$

$$E\left(e^{jw\delta_1(t,s)}\right) = P(t-s < T) \cdot e^{jw} + P(t-s > T) \cdot 1 = e^{-u(t-s)} e^{jw} + 1 - e^{-u(t-s)}$$

$$\Rightarrow \phi_{Y_2(t)}(w) = \exp\left(\frac{\lambda}{u}\left(1 - e^{-ut}\right)\left(e^{jw} - 1\right)\right) \Rightarrow Y_2(t) \sim \text{泊松分布 } \frac{\lambda}{u}\left(1 - e^{-ut}\right)$$

$$E(Y_2(t)) = \frac{\lambda}{u}\left(1 - e^{-ut}\right)$$

② $N \sim$ 泊松分布 γ：$Y_1(t) \sim$ 泊松分布 γe^{-ut}

$$\Rightarrow Y(t) = Y_1(t) + Y_2(t) \sim \text{泊松分布}\left(\gamma e^{-ut} + \frac{\lambda}{u}\left(1 - e^{-ut}\right)\right)$$

e.g.4 (24T6) $T \sim \text{Exp}(u)$，$N \sim$ 泊松分布 α，$a(t)$ 正在，$c(t)$ 已完成

(1) $a(t) \sim$ 泊松分布 $\left(\alpha e^{-ut} + \lambda\alpha\left(1 - e^{-ut}\right)\right)$。不是 Poisson 过程 (少事件：顾客十分)

(2) $c_1(t) \sim \text{Poisson}\ u\left(1 - e^{-ut}\right)$，$c_2(t) = \sum_{k=1}^{N(t)} c_k(t, S_k)$，$c_k(t, S_k) = \begin{cases} 1 & t - S_k \geq T \\ 0 & \end{cases}$

$\Rightarrow c_2(t) \sim \text{Poisson}\ \lambda t - \lambda\alpha\left(1 - e^{-t/\alpha}\right)$，$c(t) \sim \text{Poisson}(\cdot\cdot)$。

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总结：设服务时间 $T \sim \exp(\mu_T)$，开始前人数为 M，开始后人数为 $N(t)$，

正在服务为 $Q(t) = Q_1(t) + Q_2(t)$，已完成为 $C(t) = C_1(t) + C_2(t)$

(1) $M = \text{const}$：$Q_1(t) \sim B(M, e^{-\mu_T t})$，$Q_2(t) \sim P_o\left(\frac{\lambda}{\mu}(1-e^{-\mu t})\right)$

$C_1(t) \sim B(M, 1-e^{-\mu_T t})$，$C_2(t) \sim P_o\left(\lambda t - \frac{\lambda}{\mu}(1-e^{-\mu t})\right)$

(2) $M \sim Poisson(\mu_M)$：$Q_1(t) \sim P_o(\mu_M e^{-\mu_T t})$

$C_1(t) \sim P_o(\mu_M(1-e^{-\mu_T t}))$

注：若已知过程为 Poisson，直接算期望来得到参数，比算中快

形如 $\xi(t)=\sum_{i=1}^{N(t)} X_i h(t-s_i)$ 形式的过滤Poisson，有以下结论

① $E(\xi(t))=\lambda E(X)\int_0^t h(\tau)d\tau$

② $Var(\xi(t))=\lambda E(X^2)\int_0^t h^2(\tau)d\tau$

③ $R_\xi(\tau)=\lambda E(X^2)\, h(\tau)*h(-\tau)+E^2(\xi(t))$

④ $S_\xi(\omega)=\lambda E(X^2)|H(\omega)|^2+2\pi E^2(\xi(t))\,\delta(\omega)$

⑤ $\ln\Phi_\xi(\omega)=\lambda(\Phi_X(\omega)-1)$，$\ln\Phi_\xi(\omega)=\lambda\int_0^t[\Phi_X(\omega\cdot h(\tau))-1]d\tau$

} h不因果要求且有非平稳项；要求 $t\to\infty$（进入稳态）

证①：$E(\xi(t))=\lambda E(X)\int_0^t h(t-s)ds=\lambda E(X)\int_0^t h(\tau)d\tau$

② $Var(\xi(t))=E(\xi^2(t))-E^2(\xi(t))$，$E(\xi^2(t))=E\left(E(\xi^2(t)\mid N(t)=n)\right)$

$E(\xi^2(t)\mid N(t)=n)=E\left[\left(\sum_i X_i h(t-s_i)\right)\left(\sum_j X_j h(t-s_j)\right)\right]$

$i=j$：$=E(X^2)\int_0^t h^2(t-s)\frac{1}{t}ds=\frac{1}{t}E(X^2)\int_0^t h^2(\tau)d\tau$　　共 n 项

$i\neq j$：$\because X_i$ 与 X_j 独立，s_i 与 s_j 独立（在 $|N(t)=n$ 条件下）$\therefore=E^2(X)\left(\int_0^t h(\tau)\frac{1}{t}d\tau\right)^2$　　$n(n-1)$ 项

$\Rightarrow E(\xi^2(t)\mid N(t)=n)=\frac{n}{t}E(X^2)\int_0^t h^2+\frac{n(n-1)}{t^2}E^2(X)\left(\int_0^t h\right)^2$

$\Rightarrow Var(\xi(t))=\lambda E(X^2)\int_0^t h^2+\frac{(\lambda t)^2+\lambda t-\lambda t}{t^2}E^2(X)\left(\int_0^t h\right)^2-\left(\lambda E(X)\int_0^t h\right)^2$

$=\lambda E(X^2)\int_0^t h^2(\tau)d\tau$

---

③ $R_\xi(\tau)=E(\xi(t)\xi(t+\tau))=E\left(E(\xi(t)\xi(t+\tau)\mid N(t+\tau)=m)\right)$ 设h因果，$\therefore$ $\xi(t)$ 的求和上限也可写为m

$E(\xi(t)\xi(t+\tau)\mid N(t+\tau)=m)=E\left[\left(\sum_{i=1}^{m} X_i h(t-s_i)\right)\left(\sum_{j=1}^{m} X_j h(t+\tau-s_j)\right)\right]$

$i=j$：$=E(X^2)\int_0^{t+\tau} h(t-s)h(t+\tau-s)\frac{1}{t+\tau}ds$　　共 m 项

$i\neq j$：$=E^2(X)\left(\int_0^{t+\tau} h(t-s)\frac{1}{t+\tau}ds\right)\left(\int_0^{t+\tau} h(t+\tau-s)\frac{1}{t+\tau}ds\right)$　　共 $m(m-1)$ 项

$\Rightarrow R_\xi(\tau)=\lambda E(X^2)\int_0^{t+\tau} h(t-s)h(t+\tau-s)ds+\lambda^2E^2(X)\left(\int_0^{t+\tau} h(t-s)ds\right)\left(\int_0^{t+\tau} h(t+\tau-s)ds\right)$

利用 $s>t$ 时 $h(t-s)=0$，$R_\xi(\tau)=\lambda E(X^2)\int_0^t h(t-s)h(t+\tau-s)ds+E(\xi(t))E(\xi(t+\tau))$

$\int_0^t h(t-s)h(t+\tau-s)ds=\int_0^t h(v)h(v+\tau)dv$

$\Rightarrow$ **$R_\xi(\tau)=\lambda E(X^2)\int_0^t h(v)h(v+\tau)dv+E(\xi(t))E(\xi(t+\tau))$**

$\xrightarrow{t\to\infty}\lambda E(X^2)\,h(\tau)*h(-\tau)+E^2(\xi)$

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e.g.1 (24T1) $\xi(t)=\sum_{i=1}^{N(t)} X_i\,\delta(t-t_i)$, $X_i \overset{i.i.d}{\sim} \begin{pmatrix} -1 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

(1) $E(\xi(t))=0$

(2) $R_\xi(\tau)=\lambda E(X^2)\,\delta(\tau)*\delta(-\tau)+0=\lambda E(X^2)\,\delta(\tau)=\lambda\delta(\tau)$

(3) $S_\xi(\omega)=\lambda$

(4) $\mathrm{Var}(\xi(t))=\lambda E(X^2)\int_0^t \delta^2(\tau)\,d\tau=\infty$

(5) $\phi_X(\omega)=E(e^{j\omega X})=\frac{1}{2}(e^{-j\omega}+e^{j\omega})=\cos\omega$

$\Rightarrow \phi_\xi(\omega)=\exp(\lambda(\cos\omega-1))$ δ可直接用 $\exp(\lambda(\phi_X(\omega)-1))$

e.g.2 (24T2) $\xi(t)=\sum_{i=1}^{N(t)} X_i\,e^{-k(t-t_i)}\,U(t-t_i)$, X_i Gauss, $E(X_i)=0$, $E(X_i^2)=1$

(1) $N(t)$ 随机 ∴ 不 Gauss

(2) $E(\xi(t))=0$, $S_\xi(\omega)=\lambda E(X^2)\,|\tilde{H}(\omega)|^2=\lambda|\tilde{H}(\omega)|^2$

$h(t)=e^{-kt}U(t) \Rightarrow \tilde{H}(\omega)=\frac{1}{k+j\omega} \Rightarrow S_\xi(\omega)=\frac{\lambda}{k^2+\omega^2}$

$S_Y(\omega)=S_\xi(\omega)\cdot|H(j\omega)|^2=\frac{\lambda}{k^2+\omega^2}\cdot\frac{\alpha^2+\omega^2}{\beta^2+\omega^2}$

(3) 输出在 0 时刻独立 ⇔ 输出为白噪声 $\Rightarrow S_Y(\omega)=\text{const} \Rightarrow |H(j\omega)|^2=C(\omega^2+k^2)$

(4) $\mathrm{Var}(\xi(2t)\mid \xi(t)=0)$

$\xi(2t)=\left(\sum_{t_i\le t}+\sum_{t<t_i\le 2t}\right) X_i\,e^{-k(2t-t_i)}$

$=e^{-kt}\underbrace{\sum_{t_i\le t} X_i\,e^{-k(t-t_i)}}_{\xi(t)}+\sum_{t<t_i\le 2t}=e^{-kt}\xi(t)+\boxed{\xi_{new}(t)}$

⇔ $(0,t]$ 的 Poisson

$\Rightarrow \xi(2t)\mid\xi(t)=0 \overset{②}{=} 0+\xi_{new}(t)=\xi_{new}(t)$

$\mathrm{Var}(\xi_{new}(t))=\lambda E(X^2)\int_0^t h^2(\tau)\,d\tau=\lambda\int_0^t e^{-2k\tau}\,d\tau=\frac{\lambda}{2k}(1-e^{-2kt})$

(5) $E(\sin(\xi(2t)\mid\xi(t)=0))=E(\sin(\xi_{new}(t)))=\int_{-\infty}^{\infty} x\,f_\xi(x)\,dx$

∵ X_i 对称 $\Rightarrow f_\xi$ 对称 $\Rightarrow \int_{-\infty}^{\infty} x f_\xi(x)\,dx=0$

奇 偶

式用 $E(\sin(\omega\xi(t)))=\mathrm{Im}(\phi_\xi(\omega))$

$\phi_\xi(\omega)$ 实 ∴ $=0$

①